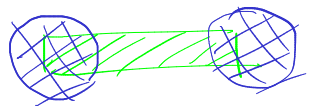
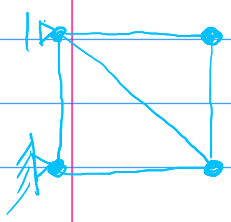
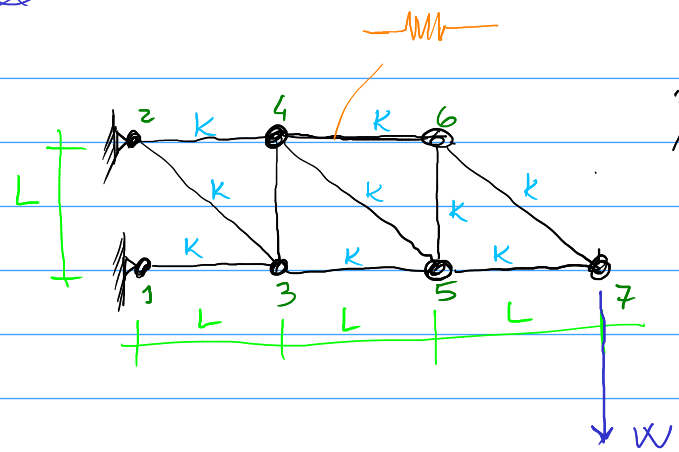
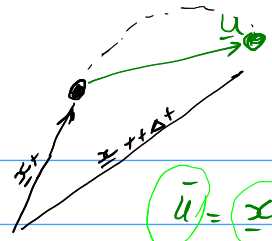


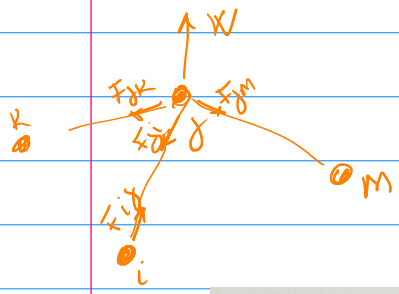


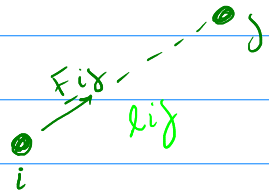


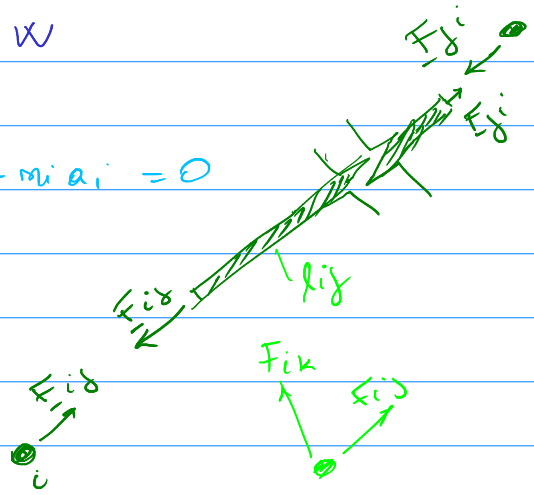


$$\bar{u} = \underline{x}_{t+\Delta t} - \underline{x}_t$$

$$\sum_{j=1}^{N} \underline{F}_{ij} = m_i \underline{a}_i \Rightarrow \sum_{j=1}^{N} \underline{F}_{ij} - m_i a_i = 0$$

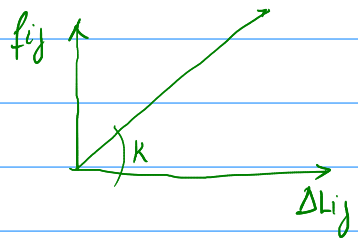






$$\begin{cases} \sum_{j=1}^{N} \underline{F}_{1j} = m_1 \underline{a}_1 \\ \sum_{j=2}^{N} \underline{F}_{2j} = m_2 \underline{a}_2 \\ \vdots \\ \sum_{j=3}^{N} \underline{F}_{7j} = m_7 \underline{a}_7 \end{cases}$$

$$\underline{F}_{ij} = -\underline{F}_{ji}$$

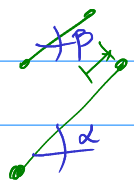


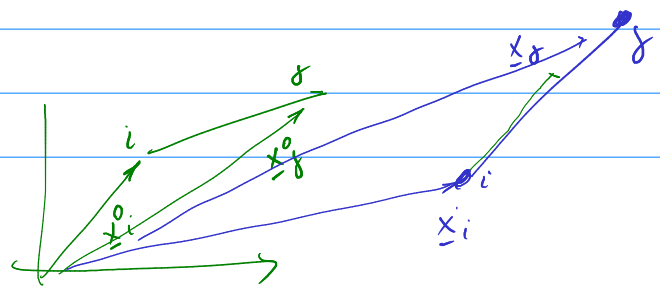
scalar con signo

Modello contitutivo

$$f_{ij} = k_{ij} \cdot \Delta l_{ij}$$

$$f_{ij} = k_{ij} \left(\| \underline{x}_j - \underline{x}_i \| - \| \underline{x}_j^0 - \underline{x}_i^0 \| \right)$$

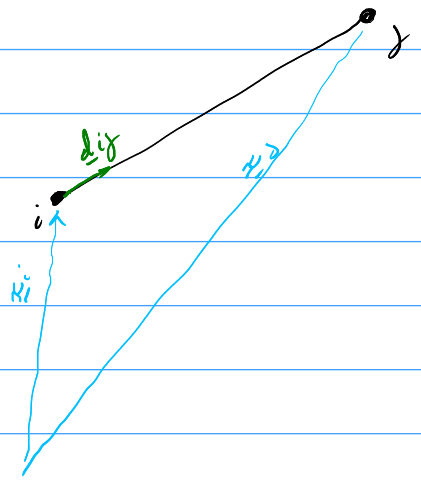




$f_{ij} > 0$ si está traccionado

$f_{ij} = 0$ si no tiene deformación (cambio de longitud)

$f_{ij} < 0$ si está comprimido



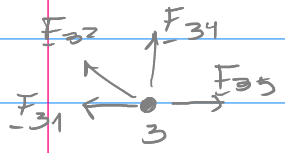
$$\underline{d}_{ij} = \frac{\overbrace{\underline{x}_j - \underline{x}_i}^{\vec{ij}}}{\|\underline{x}_j - \underline{x}_i\|}$$

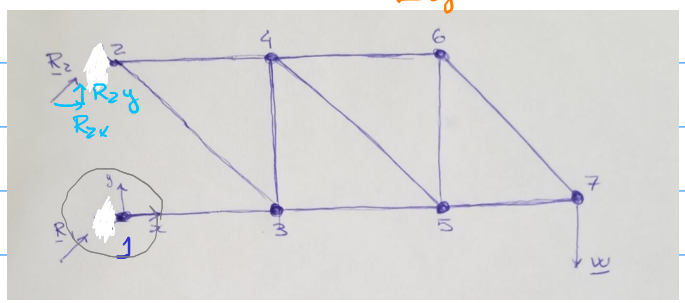
↳ vector dirección

Posición deformación

$$\underline{d}_{ij} = \frac{\underline{x}_j^0 - \underline{x}_i^0}{\|\underline{x}_j^0 - \underline{x}_i^0\|}$$

$$\underline{F}_{ij} = f_{ij}\,\underline{d} = k_{ij}\left(1 - \frac{\overbrace{\|\underline{x}_j^0 - \underline{x}_i^0\|}^{L_{ij}^0}}{\underbrace{\|\underline{x}_j - \underline{x}_i\|}_{L_{ij}}}\right)\cdot(\underline{x}_j - \underline{x}_i)$$

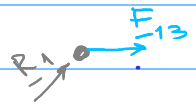




$$\dot{\underline{v}}_1 = \frac{d\underline{v}_1(t)}{dt}$$

$$\underline{a}_1 = \dot{\underline{v}}_1 \qquad \underline{v}_1 = \dot{\underline{x}}_1$$

$$\underline{a}_1 = \ddot{\underline{x}}_1$$



(1) $\sum \underline{F}_{1j} = \cancel{\underline{F}_{12} + \underline{R}_1} = m_1 \cdot \underline{a}_1 = m_1 \cdot \dot{\underline{v}}_1$

(2) $\sum \underline{F}_{2j} = \underline{F}_{24} + \underline{F}_{23} + \cancel{\underline{R}_2} = m_2 \cdot \underline{a}_2 = m_2 \cdot \dot{\underline{v}}_2$

(3) $\sum \underline{F}_{3j} = \underline{F}_{35} + \underline{F}_{31} + \underline{F}_{32} + \underline{F}_{34} = m_3 \cdot \underline{a}_3 = m_3 \cdot \dot{\underline{v}}_3$

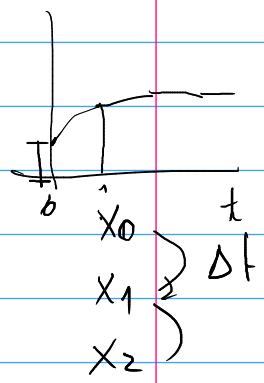
(4) $\sum \underline{F}_{4j} = \underline{F}_{46} + \underline{F}_{45} + \overbrace{\underline{F}_{43}}^{-\underline{F}_{34}} + \underline{F}_{42} = m_4 \cdot \underline{a}_4 = m_4 \cdot \dot{\underline{v}}_4$

(5) $\sum \underline{F}_{5j} = \underline{F}_{57} + \underline{F}_{56} + \underline{F}_{53} + \overbrace{\underline{F}_{54}}^{-\underline{F}_{45}} = m_5 \cdot \underline{a}_5 = m_5 \cdot \dot{\underline{v}}_5$

(6) $\sum \underline{F}_{6j} = \underline{F}_{64} + \underline{F}_{67} + \underline{F}_{65} = m_6 \cdot \underline{a}_6 = m_6 \cdot \dot{\underline{v}}_6$

(7) $\sum \underline{F}_{7j} = \underline{F}_{76} + \underline{F}_{75} + \underline{w} = m_7 \cdot \underline{a}_7 = m_7 \cdot \dot{\underline{v}}_7$

No sirven para determinar reacciones



(8) $v_7 = \dot{x}_7$

(9) $v_5 = \dot{x}_5$

(10) $v_6 = \dot{x}_6$

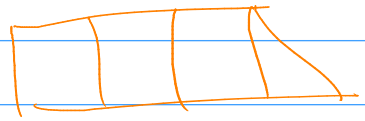
(11) $v_4 = \dot{x}_4$

(12) $v_3 = \dot{x}_3$

(13) $v_2 = \dot{x}_2$

(14) $v_1 = \dot{x}_1$

$$\boxed{\text{Incógnitas} \quad \underline{v}(t), \ \underline{x}_i(t)}$$



Condiciones iniciales

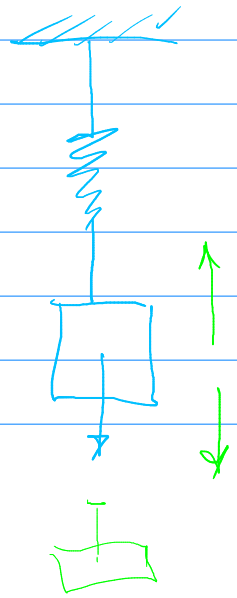
$$\underline{x}_i(t=0) = \underline{x}_i^0 \quad \forall i$$

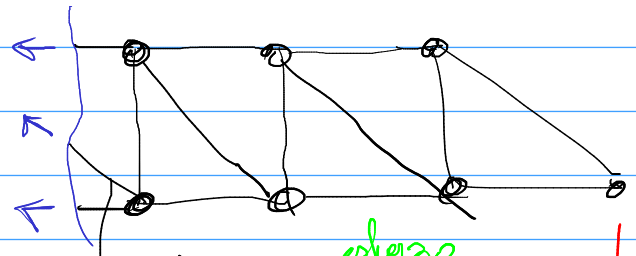
$$\underline{v}_i(t=0) = \underline{v}_i = 0 \quad \forall i$$

Condiciones de borde

$$\underline{v}_1 = 0 \qquad \underline{x}_1 = \underline{x}_1^0 \qquad \forall t$$

$$\underline{v}_2 = 0 \qquad \underline{x}_2 = \underline{x}_2^0 \qquad \forall t$$





esfuerzo estático vs esfuerzo dinámico

KJ

Programa

$$\begin{cases} \Sigma F_x = 0 \\ \Sigma F_y = 0 \\ \Sigma M = 0 \end{cases}$$